

TABLE II
Notable Printers Manufactured Outside India Under ₹ 100,000

Company	Printer	Features	Price	Warranty
RS Components & Controls Ltd	IdeaWerk 	- Build volume of 15.2x15.2x15.2cm (6x6x6-inch) with a precision of 0.18mm layer thickness - Uses non-toxic, recyclable PLA material - Has offline printing option that avoids the need to connect the printer to the PC - Can start by loading your print files in an SD card and giving print instructions from a touchscreen interface	₹ 75,000*#	One Year
XYZ Printing	da Vinci 1.0 AIO 	- Resolution: fine 0.1mm (100 microns), standard 0.2mm (200 microns) - Speed: 0.3mm (300 microns), ultra-fast 0.4mm (400 microns) - Fused filament fabrication (FFF) technology - Scanner technique - slit laser triangulation	₹ 77,400*#	One Year
RepRap	Prusa i3 complete kit 	- Open source RepRap 3D printer 20.3cm (8-inch) cubed build volume allows for large-scale functional 3D prints - Makes use of 3D-printed parts, and many replacement, upgrade and customisation parts can be printed once the kit is assembled	₹ 47,700*#	—
Solidoodle	Solidoodle 4 	- Provides 20.3x20.3x20.3cm (8x8x8-inch) heated build plate with sleek black powder-coated enclosure and ABS door with window - Creates plastic parts up to 20.3x20.3x20.3cm (8x8x8-inch) - Resolution up to 0.1mm, uses 1.75mm plastic filament (ABS/PLA), 34.3x35.5x38.1cm (13.5x14x15-inch) case footprint (LxWxH)	₹ 66,500*#	60 days
WobbleWorks	3Doodler Pen 2.0 	- Freedom to create any design, any object, at any time, no need for software or any technical design ability - Extrusion speed control, LED notification when extruder is hot, depending on the type of filament being used, 12.7cm (5-inch) power cord and 127x25.4cm (50x10-inch) strands of PLA filament	₹ 9200*#	One Year

*Price includes taxes and shipping charges; #Available for purchase via distributors in India; ** Price excluding taxes and shipping charges to India

Auto bed levelling is a comforting feature to consider

In India, out of all 3D printer makers, only a handful have this functionality. Before the process of printing starts, auto levelling makes the printer inspect the actual condition of the bed at various points and understand the prevailing situation of the bed. Even if the bed is not levelled, the printer adjusts to the bed position accordingly and prints the object flawlessly.

A 3D printer has two main parts to it: printing head and printing bed. Head is where the filament gets melted and deposited on the surface called the bed. The head moves in x and y directions and the bed moves in z direction. You can hardly find any printer in this category and price range, where the printing bed moves on all three axes.

—Vishal J.C., CEO, Grey Sim Ltd

Considering this price range, differences in 3D printers available in the market are with respect to parameters such as build volume, price, number of extruders or underlying chips that make movements faster (speed of the printer).

Coming to technological advancements, the fundamental approach remains the same, especially for printers under ₹ 100,000. Nitin Gandhi, co-founder, LBD Makers, says, "For printers within this price range, recent notable additions have been Wi-Fi connectivity, auto bed levelling, Bluetooth connectivity and dual nozzles."

Upputuri says, "There is a rise in the speed and number of extruders for a 3D printer." He adds, "Earlier dual extruder printers were costly but today these are available for less than ₹ 100,000. This is mostly because of the increase in volume and stabilisation of FDM."

Apart from automated bed levelling and wireless connectivity being embedded into 3D printers, Nikhil Chowdhary, technical head, Design & Prototyping Centre, ESCI - The Institution of Engineers (India), says, "Online support, all-metal nozzle, partially-enclosed build chambers

and portability/collapsible construction are some other features provided by manufacturers. Modular 3D printers are encouraging customisation. Also, direct light processing (DLP) technology is finding a new foothold and will be a strong competitor for FDM only in jewellery making and micro-electronics prototyping."

He adds, "Some manufacturers are introducing accessories like acrylonitrile-butadiene-styrene (ABS) spray for bed preparation and spray to smoothen printed parts, along with 3D printers."

Locked or open printers—the choice is yours

Broadly speaking, FDM printers available in the market can be divided into two groups: locked 3D printers and open 3D printers, informs Reddy.

In locked 3D printers, the whole printer is a locked system and the customer has to purchase the same company's filaments, whereas in the case of open 3D printers, a buyer can pur-